

LINEAR AND ROBUST CONTROL SYSTEMS

ELEC 9731

4B - Linear State Feedback

© Prof. Victor Solo

School of Electrical Engineering and
Telecommunications

UNSW

Sydney

Slide 1

TOPICS

- 1** Linear State Feedback (LSFB)
- 2** Bass-Gura Formula
- 3** Modal View of LSFB
- 4** Disturbance Rejection
- 5** minimal Systems

LINEAR STATE VARIABLE FEEDBACK

Since states are often position, velocity, acceleration, it is a natural idea to feedback a linear combination of them.

We pursue this idea in two stages:
first suppose all states are measured, then
discuss estimating the states

- SS model

$$\dot{x} = Ax + bu, y = c^T x$$

- LSFB

$$u = r - k^T x$$

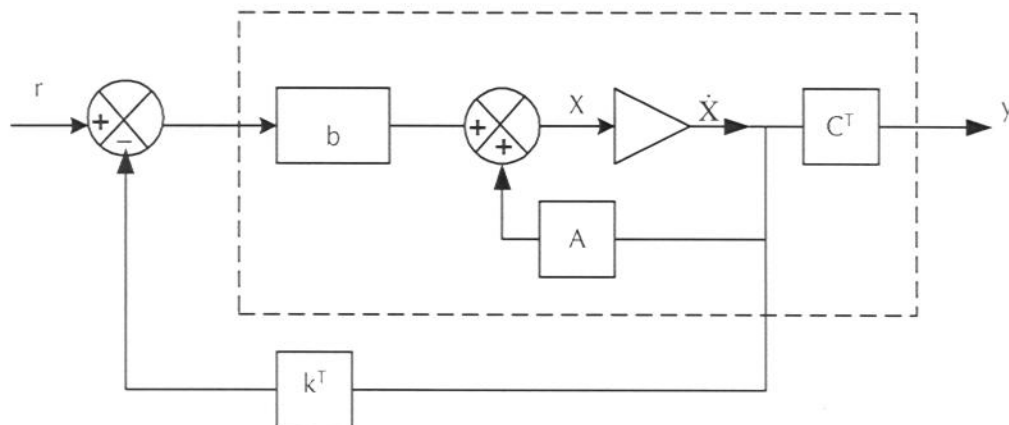


FIGURE 4.6 LS FB
Dotted lines show whole plant

LINEAR STATE VARIABLE FEEDBACK

- Continued

Aim now is to gain understanding of LSFB from a LT point of view. Develop closed loop behaviour using Laplace transforms (neglect initial conditions)

$$\begin{aligned}
 s\bar{x} &= A\bar{x} + b\bar{u} \\
 \bar{u} &= \bar{r} - k^T\bar{x} \\
 \Rightarrow s\bar{x} &= A\bar{x} + b(\bar{r} - k^T\bar{x}) \\
 &= (A - bk^T)\bar{x} + b\bar{r} \\
 \Rightarrow \bar{x} &= (sI - A + bk^T)^{-1}b\bar{r} \\
 \Rightarrow \bar{y} &= c^T\bar{x} \\
 &= c^T(sI - A + bk^T)^{-1}b\bar{r}
 \end{aligned}$$

So

$$G_k(s) = CLTF = c^T(sI - A + bk^T)^{-1}b$$

c.f. OLTF

$$P(s) = c^T(sI - A)^{-1}b = \frac{b(s)}{a(s)}$$

Slide 4

LSFB - CLTF

To get to polynomial form, use the matrix inversion lemma (MIL)

$$\begin{aligned} (sI - A + bk^T)^{-1} &= (sI - A)^{-1} \\ &\quad - \frac{(sI - A)^{-1}bk^T(sI - A)^{-1}}{1 + k^T(sI - A)^{-1}b} \end{aligned}$$

The CLTF is then

$$\begin{aligned} G_k(s) &= c^T(sI - A + bk^T)^{-1}b \\ &= P(s) - \frac{P(s)k^T(sI - A)^{-1}b}{1 + k^T(sI - A)^{-1}b} \\ &= \frac{P(s)}{1 + k^T(sI - A)^{-1}b} \end{aligned}$$

Denote the characteristic polynomials

$$\begin{aligned} a(s) &= \det(sI - A) \\ a_k(s) &= \det(sI - A + bk^T) \end{aligned}$$

LSFB-CLTF - Continued

We can write then

$$\begin{aligned} k^T (sI - A)^{-1} b &= \frac{k^T \text{adj}(sI - A) b}{\det(sI - A)} \\ &= \frac{\bar{k}(s)}{a(s)} \end{aligned}$$

for some polynomial $\bar{k}(s)$.

On the one hand,

$$\begin{aligned} G_k(s) &= \frac{c^T \text{adj}(sI - A + bk^T)^{-1} b}{\det(sI - A + bk^T)} \\ &= \frac{b_k(s)}{a_k(s)} \end{aligned}$$

On the other hand,

$$\begin{aligned} G_k(s) &= \frac{\frac{b(s)}{a(s)}}{1 + \frac{\bar{k}(s)}{a(s)}} \\ &= \frac{b(s)}{a(s) + \bar{k}(s)} \\ &= \frac{b(s)}{a_k(s)} \end{aligned}$$

Slide 6

LSFB - DESIGN

- Zeroes Unaffected

We deduce the CLTF has the same numerator polynomial as the OLTF. So LSVFB does not affect position of zeros.

- Poles Changed

Only the denominator polynomial is changed. So CL poles are changed.

- Design=specify CL poles

To turn LSFB into a design procedure we specify the CL poles by specifying a desired CL denominator polynomial

$$a_k(s) = a_d(s)$$

Then we solve this equation for k i.e. solve

$$\begin{aligned} a(s) + \bar{k}(s) &= a_d(s) \\ \Rightarrow 1 + \frac{\bar{k}(s)}{a(s)} &= \frac{a_d(s)}{a(s)} \\ \Rightarrow 1 + k^T (sI - A)^{-1} b &= \frac{a_d(s)}{a(s)} \end{aligned}$$

LSFB - DESIGN - Continued

We solve the equation by polynomial division. as follows

$$\begin{aligned}
 (sI - A)^{-1} &= s^{-1} \left(I - \frac{A}{s} \right)^{-1} \\
 &= s^{-1} \sum_{m=0}^{\infty} \frac{A^m}{s^m} = \sum_{r=1}^{\infty} \frac{A^{r-1}}{s^r} \\
 \Rightarrow \frac{a_d(s)}{a(s)} - 1 &= \sum_{r=1}^{\infty} \frac{k^T A^{r-1} b}{s^r} \\
 &= \sum_{r=1}^{\infty} h_r s^{-r} \\
 &= \frac{a_d(s) - a(s)}{a(s)}
 \end{aligned}$$

Applying the polynomial division equation gives

$$\mathcal{A}_-^{(m)} h^{(m)} = g_d^{(m)} = [a_{d,1} - a_1, \dots, a_{d,n} - a_n]^T$$

Take $m = n$ and write this as

$$\mathcal{A}_- h = a_d - a = g_d$$

LSFB - DESIGN - Continued

Now however

$$\begin{aligned}
 h^T &= (h_1, \dots, h_n)^T \\
 &= (k^T b, k^T A b, \dots, k^T A^{n-1} b) \\
 &= k^T \mathcal{C}(A, b) \\
 \Rightarrow g_d &= \mathcal{A}_- \mathcal{C}^T(A, b) k \\
 \Rightarrow k &= [\mathcal{C}^T(A, b)]^{-1} \mathcal{A}_-^{-1} g_d
 \end{aligned}$$

This is the Bass - Gura formula

N.B.

- (i) Because $\mathcal{A}_-$ is lower triangular it is well-conditioned.
- (ii) If $\mathcal{C}(A, b)$ is ill-conditioned because of near uncontrollable modes then k will have large elements.
- (iii) Solution requires (A, b) is controllable.
- (iv) This is the pole placement theorem.
- (v) An alternative is Ackerman's formula - less general

Bass- Gura Example

Compute the LSFB gain vector for control of OL system

$$P(s) = \frac{4}{s(s+1)(s+4)} = \frac{1}{s} + \frac{-4/3}{s+1} + \frac{1/3}{s+4}$$

with desired closed loop denominator polynomial

$$a_d(s) = s^3 + 7s^2 + 6s + 4$$

First convert to SS form. There are several options ; we use modal form

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -4 \end{pmatrix}, b = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, c = \begin{pmatrix} 1 \\ -4/3 \\ 1/3 \end{pmatrix}$$

Also $a(s) = s^3 + 5s^2 + 4s$ so

$$a_d - a = 2s^2 + 2s + 4 \Rightarrow g_d = (2, 2, 4)^T$$

Bass- Gura Example Continued

We have next

$$\begin{aligned}\mathcal{A}_- &= \begin{pmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 4 & 5 & 1 \end{pmatrix} \\ \mathcal{C}(A, b) &= (b, Ab, A^2b) \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 1 & -1 & 1 \\ 1 & -4 & 16 \end{pmatrix} \\ \Rightarrow k &= [\mathcal{A}_- \mathcal{C}^T(A, b)]^{-1} g_d \\ &= \begin{pmatrix} 1 & -1.33 & 2.33 \end{pmatrix}\end{aligned}$$

LSFB - MODAL VIEW

Use the EVD for $A=P\Lambda Q^T$ to find a partial fraction expansion (recall $PQ^T = Q^T P = I$)

$$\begin{aligned}
 k^T(sI - A)^{-1}b &= k^T(sI - P\Lambda Q^T)^{-1}b \\
 &= k^T[P(sI - \Lambda)Q^T]^{-1}b \\
 &= k^T Q^{-T}(sI - \Lambda)^{-1}P^{-1}b \\
 &= k^T P(sI - \Lambda)^{-1}Q^T b \\
 &= \kappa^T(sI - \Lambda)^{-1}\beta \\
 &= \sum_1^n \frac{\kappa_r \beta_r}{s - \lambda_r}
 \end{aligned}$$

$$\text{where } \kappa = P^T k \neq P^T c$$

$$\Rightarrow \frac{a_d(s)}{a(s)} = 1 + \sum_1^n \frac{\kappa_r \beta_r}{s - \lambda_r}$$

LSFB - MODAL VIEW - Continued

Expand $a_d(s), a(s)$ in terms of roots

$$\frac{a_d(s)}{a(s)} = \frac{\prod_1^n (s - \mu_j)}{\prod_1^n (s - \lambda_j)}$$

$$\mu_j = \text{CL poles, } \lambda_j = \text{OL poles}$$

Multiply across by $s - \lambda_k$, set $s = \lambda_k$ to get

$$\kappa_k \beta_k = (\lambda_k - \mu_k) \prod_{j \neq k} \frac{\lambda_k - \mu_j}{\lambda_k - \lambda_j}$$

This is the Mayne-Murdoch formula.

Clearly as we move μ_k away from λ_k , then $\kappa_k \beta_k$ gets large.

If mode k is nearly uncontrollable so that $|\beta_k| \ll 1$, then κ_k the gain will be very large. This will be unacceptable in practice.

LSFB & DISTURBANCES

Suppose a *dc* disturbance appears in the state equation

$$\dot{x} = Ax + bu + w, y = c^T x$$

with plain LSFB the disturbance will yield a steady state error. To overcome this expand the SS model by adding a disturbance model

$$\begin{aligned} \dot{d} &= c^T x \\ \Rightarrow \begin{pmatrix} \dot{x} \\ \dot{d} \end{pmatrix} &= \begin{pmatrix} A & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ d \end{pmatrix} + \begin{pmatrix} b \\ 0 \end{pmatrix} u + \begin{pmatrix} w \\ 0 \end{pmatrix} \end{aligned}$$

Now apply LSFB to this new model

$$u = r - k^T x - k_d d, \quad (r = 0)$$

LSFB & DISTURBANCES - Continued

The CL system is now ($r = 0$)

$$\begin{pmatrix} \dot{x} \\ \dot{d} \end{pmatrix} = \begin{pmatrix} A - bk^T & -bk_d \\ c^T & 0 \end{pmatrix} \begin{pmatrix} x \\ d \end{pmatrix} + \begin{pmatrix} w \\ 0 \end{pmatrix} + \begin{pmatrix} b \\ 0 \end{pmatrix} r$$

If (k^T, k_d) produces a stable CL system, then at steady state $\dot{x} = 0$, $\dot{d} = 0$ so

$$0 = c^T x(\infty) = y(\infty) \text{ as required.}$$

LSFB Continued

Q: Does state feedback effect controllability?

A: No

Proof: Consider SS system A, b, c . After LSFB closed loop system is

$$A - bk^T, b, c$$

To enable LSFB, A, b, c must have been controllable. Suppose closed loop system is not controllable. By PBH test there is a left evec q which is orthogonal to b , i.e.

$$\begin{aligned} q^T(A - bk^T) &= \lambda q^T, q^T b = 0 \\ \Rightarrow q^T A &= \lambda q^T, q^T b = 0 \end{aligned}$$

which contradicts controllability of A, b, c

MINIMAL SYSTEMS

- We say the SS system A, b, c is minimal if A, b is controllable and c, A is observable

Theorem A, b, c is minimal iff $a(s), b(s)$ are coprime

(where $\left[c^T (sI - A)^{-1} b = \frac{b(s)}{a(s)} \right]$).

Proof. Follows from simple consideration of pole zero cancellations.

- If $a(s), b(s)$ are coprime we say $\frac{b(s)}{a(s)}$ is irreducible.

STATE FEEDBACK AND OBSERVABILITY

If A, b, c is minimal, then after LSFB ,
 $A - bk^T, b, c$ is still controllable.

But closed loop transfer function is

$$c^T(sI - A + bk^T)^{-1}b = \frac{b(s)}{a_k(s)}$$

If $b(s), a_k(s)$ are coprime

i.e. no poles of $a_k(s)$ cancel any zeros of $b(s)$

then $A - bk^T, b, c$ is still observable and so still minimal.